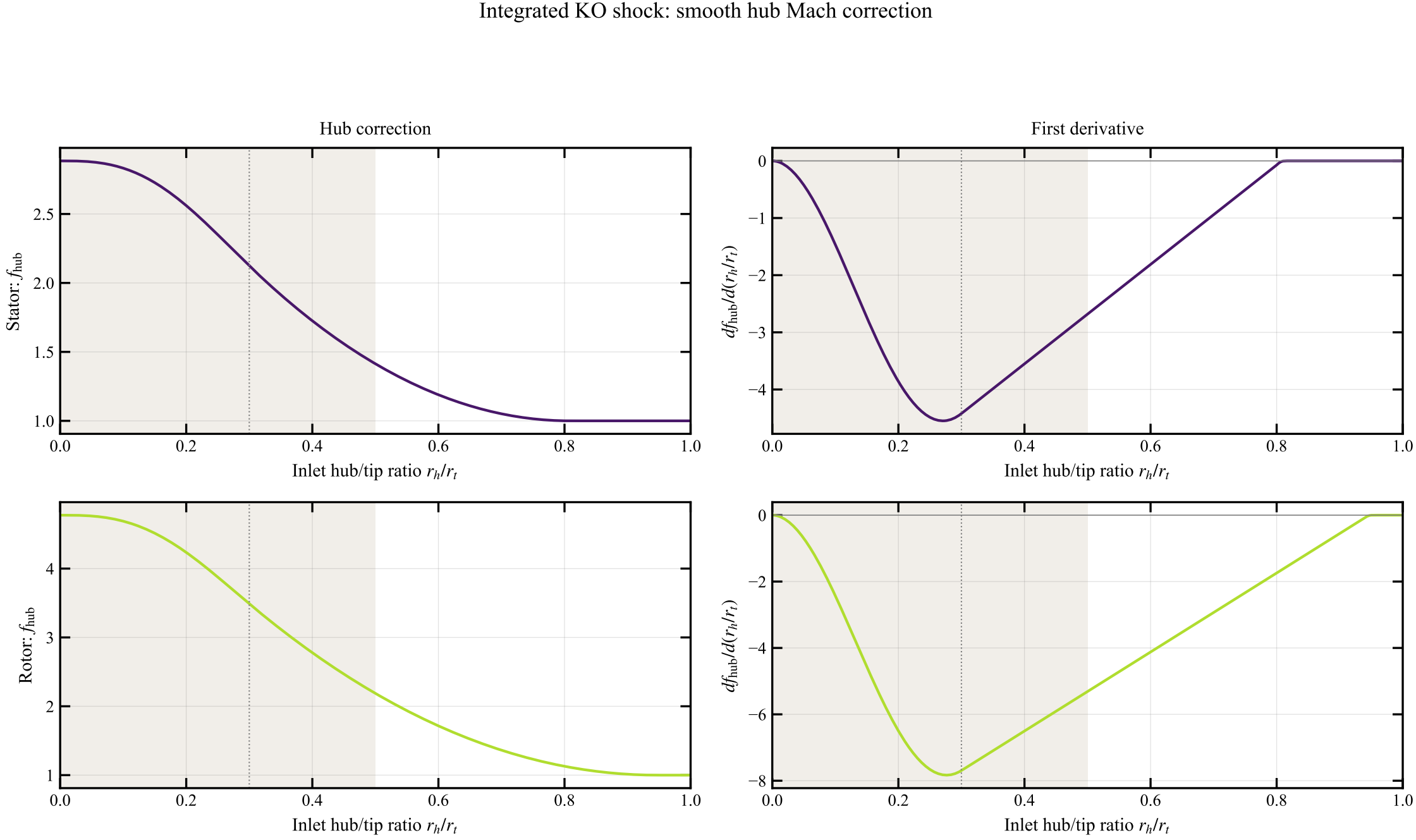
Integrated KO shock: smooth hub Mach correction
Hub correction
First derivative
Stator: f_{hub}
Rotor: f_{hub}
$df_{hub}/d(r_h/r_t)$
Inlet hub/tip ratio r_h/r_t
0.0
0.2
0.4
0.6
0.8
1.0
1.0
1.5
2.0
2.5
1
2
3
4
0
−1
−2
−3
−4
−6
−8
Tint: r_h/r_t < 0.5, beyond the digitized chart. Dotted line: flattening below 0.3.
Zero ratio is a mathematical limit for this factor; it is outside the physical shock-loss domain.